

114 學年度第二學期 離散數學 HW06

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1 題目

1.1 作業繳交說明

- 從 overleaf 下載 hw06.zip
- 從 overleaf 下載 hw06.pdf
- 建立一個新檔案夾，命名如下
- 離散數學_114590xxx_姓名_HW06
- 放入 hw06.zip
- 放入 hw06.pdf
- 將此檔案夾，壓縮成 zip 檔，檔名如下
- 離散數學_114590xxx_姓名_HW06.zip
- 將此壓縮檔，上傳到北科 i 學園 PLUS
- Do not share your homework with any living creatures.
- 作業遲交以零分計。

1.2 題號

- page 418, chapter 6.1 Exercise 36
- page 444, chapter 6.4 Exercise 10
- page 455, chapter 6.5 Exercise 14
- page 455, chapter 6.5 Exercise 16
- page 455, chapter 6.5 Exercise 20
- page 457, chapter 6.5 Exercise 64
- page 457, chapter 6.5 Exercise 68

2 作答

2.1 page 418, chapter 6.1 Exercise 36

How many functions are there from the set $\{1, 2, \dots, n\}$, where n is a positive integer, to the set $\{0, 1\}$?

There are 2^n such functions, since there is a choice of 2 function values for each element of the domain.

2.2 page 444, chapter 6.4 Exercise 10

Use the binomial theorem to expand $(3x - y^2)^4$ into a sum of terms of the form $cx^a y^b$, where c is a real number and a and b are nonnegative integers.

$$\begin{aligned}(3x - y^2)^4 &= \sum_{j=0}^4 \binom{4}{j} (3x)^{4-j} (-y^2)^j \\ &= 81x^4 + -108x^3y^2 + 54x^2y^4 + \\ &\quad -12xy^6 + y^8\end{aligned}$$

2.3 page 455, chapter 6.5 Exercise 14

How many solutions are there to the equation

$$x_1 + x_2 + x_3 + x_4 = 17,$$

where x_1, x_2, x_3 , and x_4 are nonnegative integers?

$$C(4 + 17 - 1, 17) = C(20, 17) = 1140$$

2.4 page 455, chapter 6.5 Exercise 16

How many solutions are there to the equation

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 29,$$

where $x_i, i = 1, 2, 3, 4, 5, 6$, is a nonnegative integer such that

- (a) $x_i > 1$ for $i = 1, 2, 3, 4, 5, 6$?
- (b) $x_1 \geq 1, x_2 \geq 2, x_3 \geq 3, x_4 \geq 4, x_5 > 5$, and $x_6 \geq 6$?
- (c) $x_1 \leq 5$?
- (d) $x_1 < 8$ and $x_2 > 8$?

(a) $C(6 + 17 - 1, 17) = C(22, 17) = 26334$

(b) $C(6 + 7 - 1, 7) = C(12, 7) = 7922$

(c) 179976

(d) 46942

2.5 page 455, chapter 6.5 Exercise 20

How many solutions are there to the inequality

$$x_1 + x_2 + x_3 \leq 11,$$

where x_1, x_2 , and x_3 are nonnegative integers? [Hint: Introduce an auxiliary variable x_4 such that $x_1 + x_2 + x_3 + x_4 = 11$.]

$$C(4 + 11 - 1, 11) = C(14, 11) = 364$$

2.6 page 457, chapter 6.5 Exercise 64

How many different terms are there in the expansion of

$$(x_1 + x_2 + \dots + x_m)^n$$

after all terms with identical sets of exponents are added?

Each term must be of the form $Cx_1^{n_1}x_2^{n_2}\dots x_m^{n_m}$, where the n_i 's are nonnegative integers whose sum is n . The number of ways to specify a term, then, is the number of nonnegative integer solutions to $n_1 + n_2 + \dots + n_m = n$, which by Theorem 2 is $C(m + n - 1, n)$. Note that the coefficients C for these terms can be computed using Theorem 3—see Exercise 65.

2.7 page 457, chapter 6.5 Exercise 68

How many terms are there in the expansion of

$$(x + y + z)^{100}?$$

By Exercise 64, the answer is

$$C(3 + 100 - 1, 100) = C(102, 100) = 5151$$